



SafeShare 3Tier Architecture on AWS

Capstone Portfolio / Cloud Engineer

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Scalable sharing without single failure



01

Problem

Single EC2, single disk = single point of failure. Files disappear on replacement, no HA across AZs, no scale.

02

Solution

3Tier with [ALB](#) → [ASG](#) → [S3](#). Stateless Flask on private EC2, central S3 for persistence, cron bidirectional sync every 60s.

03

Key design

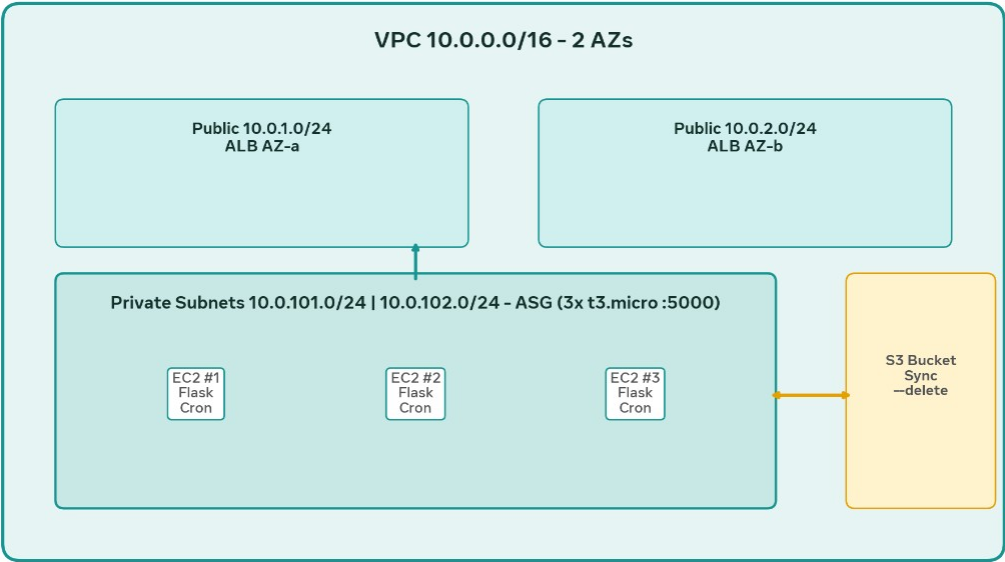
Public ALB in public subnets, private EC2 with no public IP. Secure access via IAM policy.

04

Monitoring

Athena+CUR
CloudTrail

VPC 10.0.0.0/16 across 2 AZs — public for ALB, private for compute



VPC: 10.0.0.0/16	NAT → IGW	IGW → Internet
PUBLIC AZ-A 10.0.1.0/24 ALB subnet, route to IGW, high availability		
		PUBLIC AZ-B 10.0.2.0/24 ALB subnet second AZ, HA failover
PRIVATE AZ-A 10.0.101.0/24 ASG EC2, no public IP, isolated		
		PRIVATE AZ-B 10.0.102.0/24 ASG EC2, SSM only access

S3 is the source of truth

STORAGE TIER

```
#!/bin/bash — /usr/local/bin/sync-to-s3.sh
BUCKET="safeshare-files-99"
DIR="/home/ec2-user/uploads"
LOG="/var/log/s3-sync.log"
# Bidirectional — keeps all nodes in sync
aws s3 sync $DIR s3://$BUCKET/
aws s3 sync s3://$BUCKET/ $DIR
chmod -R 777 $DIR; chown -R ec2-user $DIR
```

CRON SCHEDULE

* * * * * — every 60s

LOG FILE

/var/log/s3-sync.log



Central persistent storage

S3 survives instance replacement. No EFS cost, native versioning.



Bidirectional sync

Upload on any EC2 → S3 → all EC2s. Delete propagates with --delete flag could implement.



Fix: cronie on AL2023

dnf install cronie + systemctl enable crond --now

Private by design, secure via IAM

EC2 instances have no public IP. All access and sync is controlled by least-privilege IAM role.



No public IP

Isolated private subnets 10.0.101/102.0/24

IAM Role: SafeShare-EC2-Role

Assume: ec2.amazonaws.com • Profile: safeshare-ec2-profile



Policy 1: AWS Managed

- ✓ AmazonSSMManagedInstanceCore
- ✓ Session connect & logging
- ✓ No key pair needed



Policy 2: Custom S3

- ✓ s3:ListBucket on bucket
- ✓ s3:PutObject, GetObject
- ✓ s3:DeleteObject on /*

Upload hits ALB, lands on EC2, then cron syncs to S3

DATA FLOW

Bidirectional sync ensures every EC2 sees same files within 60 seconds.



1

User → IGW → ALB

Browser upload over 80/443. IGW routes to ALB in public [10.0.1.0/24](#) / [10.0.2.0/24](#).

2

ALB → Target Group

TG [SafeShare-TG :5000](#) health [/health](#) every 30s. Picks healthy EC2.

3

EC2 Flask save

Flask saves to [/home/ec2-user/uploads](#). Instance ID shown in UI for tracking.

4

Cron → S3 → All EC2s

***** [aws s3 sync --delete](#) both ways. Log [/var/log/s3-sync.log](#).

Highly available, persistent, and **cost optimized**

3-tier holds under replacement, scales with ASG,
and syncs via S3 within 60s — no single point of
failure.



HA across 2 AZs `10.0.1/2.0 & 101/102.0`

ALB in public subnets, EC2 in private. If AZ-a fails, AZ-b serves. Private has no public IP.



Persistent via S3 sync

Central bucket `safeshare-files-99` — bidirectional `--delete` every minute.



Self-healing & zero-downtime

ASG min 3 / desired 3 / max 6. Instance refresh replaces with launch template `t3.micro AL2023`.



Cost optimized & scalable

`t3.micro` . Scales automatically, logs at `/var/log/s3-sync.log`

● **Flask :5000 in /home/ec2-user/uploads • cron * * * * * • Health /health 30s**

3

DESIRED EC

60s

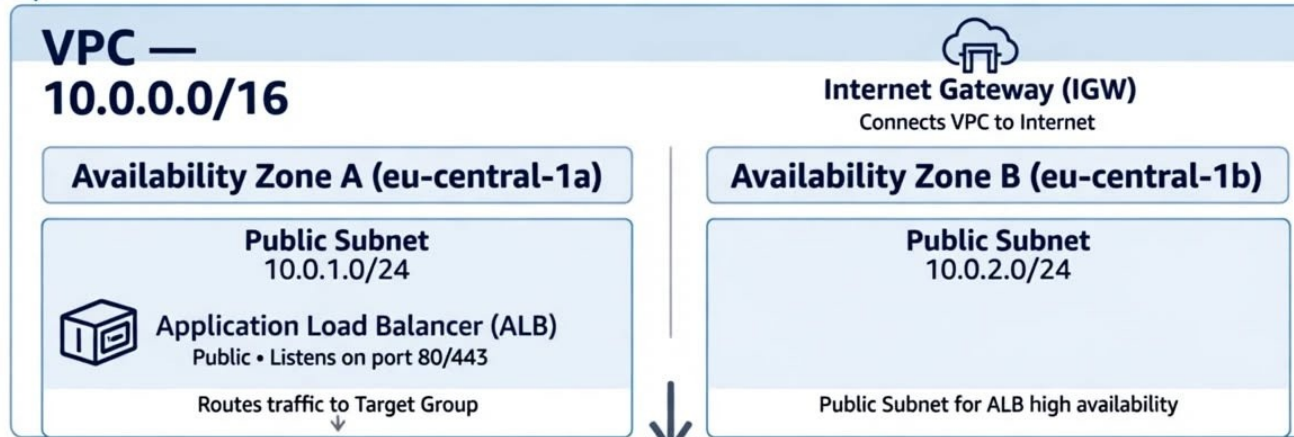
SYNC INTERVAL

SafeShare - 3Tier Architecture



VPC: 10.0.0.0/16 • AWS Region: eu-central-1

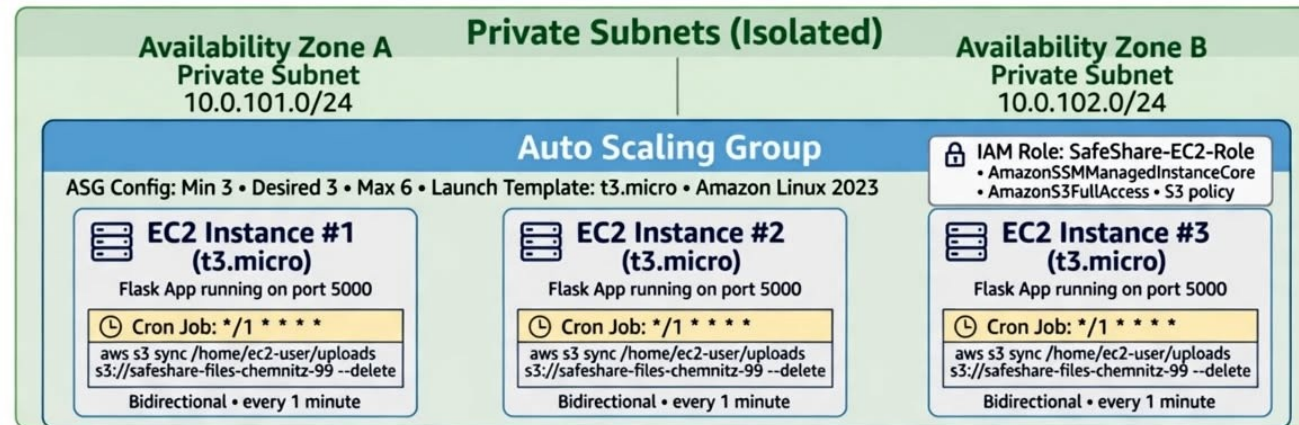
• High Availability across AZ-a & AZ-b



Protocol: HTTP • Port: 5000 • Health Check: /health every 30s



Amazon S3 Bucket
safeshare-files
Secure Object Storage • Versioning Enabled



→ Data Flow: User upload → ALB → Target Group → EC2 Instances (Flask) → Store to /home/ec2-user/uploads → S3 Sync → Replicates to all other EC2s via S3

Legend:

● **Public Subnet —**
Internet Accessible (with Route to IGW)

● **Private Subnet**
No Public IP, isolated for security

🔑 IAM Role attached to EC2 for SSM & S3 access

Key Points:

- High Availability: Resources spread across AZ-a & AZ-b
- Auto Scaling: 3 × t3.micro EC2 instances behind ALB
- Data Sync: Cron job runs every 1 min to sync uploads to S3 bidirectionally

Q&A • SAFESHARE

Thank you!

Let's scale it.



RDS for metadata



CloudFront CDN



HTTPS / ACM



S3 versioning